

Radiological Incident System using Augmented Reality (RISAR)

Muhannad Ismael* Roderick McCall† Maël Cornil‡ Mike Griffin§ Joan Baixauli¶

IT for Innovative Services (ITIS) Department
Luxembourg Institute of Science and Technology (LIST), Esch sur Alzette, Luxembourg



Figure 1: RISAR tool for CBRN events management, demonstrating scenario via head-worn and handheld devices. Middle figure is captured from IOS device. The right figure is derived from Microsoft HoloLens.

ABSTRACT

This paper presents an Augmented Reality (AR) solution called RISAR that allows for the real-time visualisation of Radiological hazards based on sensor data captured from detectors as well as Unmanned Aerial and Ground Vehicles. RISAR improves safety for first responders during radiological events by enhancing their situation awareness. This lowers the risk of harm, and with it any health impacts and costs.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Mixed / Augmented reality—

1 INTRODUCTION

Managing CBRN incidents is complex and requires an understanding potential risks to the public and the first responders. The management of incidents involves numerous first responder organisations, and potentially the military and other organisations. Those involved require accurate, real-time information on the risks and also the positioning and use of assets to detect and identify sources. However, despite advances in sensor technologies and procedures for managing such events, problems remain with existing systems [1] that are: 1) Cordons are not moved in “real-time”: cordons are placed to indicate the hot, warm and cold zones [2]. Each time there is an update to the sensor data there may be a need to manually move the cordons, this is time-consuming and may increase the exposure time to the operatives. 2) Real-time risks are not always visible: radiation is by nature invisible, however providing additional data about the position of radiation sources may be useful to operatives. 3) Low common operational picture: operatives are not always around common information displays, instead individual operatives may be using different devices. While this information may be reported back to the command post/centre, it means that not everyone has access to all the information relevant to them all the time. These issues potentially undermine the ability of first responders to manage radiological incidents.

To overcome these problems, this paper presents RISAR which al-

lows for: 1) Dynamic visualization of risks and cordons 2) Real-time data from UAVs, UGVs, handheld detectors and other sensors 3) Shared common operational picture.

2 RELATED WORK

This paper investigates system integration, data management and outdoor localisation the RISAR application. Work by [5] proposed an AR application to support urban operations. The idea is to integrate an heads-up display of chemical, biological, radiological, explosive, and other sensor data into a mobile AR system. The application is based on Battlefield Augmented Reality System (BARS) which was mainly composed of Sony Glasstron LDI-D100B, Ashetech GG24-Surveyor, InterSense, InertiaCube2, etc. The system suffers from inaccurate outdoor location since the tracking methods is only based on GPS receiver. Simultaneous Localisation and Mapping (SLAM) for indoor and outdoor using only camera tracking system for augmented reality illustrated via PTAM (Parallel Tracking and Mapping) [3].

A hybrid system using Real-Time kinematic positioning (RTK) with SLAM [4] is exploited to obtain high precision outdoor tracking using Microsoft HoloLens. The proposition is close to our suggestion in this paper. However, RISAR exploits a novel framework to achieve high accurate outdoor tracking without RTK system. The framework uses at the beginning a reference position derived from GPS system to map real and virtual coordinates as described in detail in next section.

3 SYSTEM

RISAR focusses on improving situation awareness during a radiological incident. In doing so, by allowing first responders to perceive, comprehend and plan appropriate courses of action by providing real-time accurate information. This should allow operatives reduce their overall dose radiological dose absorption and to manage the incident more effectively.

RISAR supports two modes of working, one uses live data during a real incident for example from UAVs, UGVs and handheld detectors. In addition, an emulation environment was created which avoids the need for real radioactive sources. The latter is the version discussed in this paper, as it is the one more thoroughly tested. Although the system has also been used with real radioactive sources in several countries. Early prototypes of RISAR client (i.e. the tool used by the operatives) were built using IOS and ran on an iPad. This had some limitations, namely using a handheld device as not suitable for all users. Firstly, there are ergonomics issues associated with holding a device for prolonged periods of time. Second, ma-

*e-mail: muhannad.ismael@list.lu

†e-mail: roderick.mccall@list.lu

‡e-mail: mael.cornil@list.lu

§e-mail: mikegriffin@risarx.com

¶e-mail: joan.baixauli@list.lu

nipulating a handheld device is problematic when using protective gloves. Therefore, once the technology had improved the decision was taken to migrate to Microsoft HoloLens 2. However, it was noted that for some incident staff, in particular those related outside of the safety cordon a tablet is sometimes preferred over an Head-Mounted Display (HMD). RISAR consists of three components: the server application, emulator and an AR application. AR application is conducted with a scene emulator via the server application (see figure 2). The RISAR emulator allows for the simulation of sensors (e.g., handheld detectors) and cordons with no real incident or trial being required. It permits to setup three types of cordons: red, yellow and green according to the radiation level. In additional devices such as tablets and phones play the role of handheld detectors, UGVs and UAVs. As these are moved around the emulator they are displayed in the augmented reality application. Hence, RISAR is a component within a suite of tools including: sensors, drones and command and control environments. Provided data is sent in the correct format, new devices (e.g. sensors) can be easily added.

Keyhole Markup Language (KML) messages (e.g., sensor readings) are sent from UGVs, UAVs, handheld detectors, etc. to the server application. This relays data to the AR application which displays the relevant information on AR devices. Sockets are used to allow for multiple simultaneous messages between the various system components. The server transmits from "Position and Direction Application" (see figure 2) GPS positions and Actual Absolute Heading θ defined later in this section (via dedicated WiFi) to the client application in case that AR devices don't contain GPS chips like Microsoft HoloLens, so that the client can update the position of the user relative to the virtual environment. However, updating the user position and orientation regularly using GPS information leads jittering in information display such that small errors accumulate and lead to a clear misregistration of virtual data. Therefore, we propose to use a hybrid GPS and SLAM outdoor framework with reference position. RISAR initializes the AR application at the beginning with reference position which contains a single 6-DOF position (GPS measurement). The AR application computes the distance δ in a world coordinate frame between virtual objects P_v^{world} which represent UGVs, UAVs, handheld detectors, cordon, etc. and reference positions P_{ref}^{world} which refers to the initial user location. Afterwards, the user's heading called Relative Bearing γ is calculated by subtracting the Bearing angle β from Actual Absolute Heading θ which refers to the compass direction (see figure 3). Furthermore, Bearing angle β indicates the angle between the reference and UGVs, UAVs, handheld detectors, cordon, etc. positions according to north direction. The direction angle α between $P_{ref}^{hololens}$ and $P_v^{hololens}$ in hololens coordinate frame is calculated by adding Relative Bearing γ to ϕ which represents the rotation of the HoloLens camera around y-axis. Once α is computed, the position of virtual object $P_v^{hololens}$ in the hololens coordinate frame is written as follows:

$$P_v^{hololens} = P_{ref}^{hololens} + P_{\delta}^{hololens} \quad (1)$$

$$P_{\delta}^{hololens} = \begin{pmatrix} 0 \\ 0 \\ \delta \end{pmatrix} \cos(\alpha) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \delta \end{pmatrix} (1 - \cos(\alpha)) + \begin{pmatrix} 0 \\ 0 \\ \delta \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \sin(\alpha) \quad (2)$$

$$\gamma = \beta - \theta, \alpha = \gamma + \phi \quad (3)$$

Where $P_{\delta}^{hololens}$ refers to the position after the rotate of point $(0,0,\delta)^T$ by the angle α degrees around the y-axis. Hence, the AR application transforms the GPS coordinates of handheld detectors, cordons, UGVs, UAVs, etc. derived from KML file into the SLAM camera coordinate using the reference position $P_{ref}^{hololens}$ and

Actual Absolute Heading θ which updates direction angle α . Therefore, RISAR doesn't require updating of the user position in each frame, thus avoiding noisy information, and it there provides more stable information.

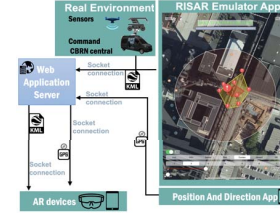


Figure 2: RISAR: schema general of the proposed system

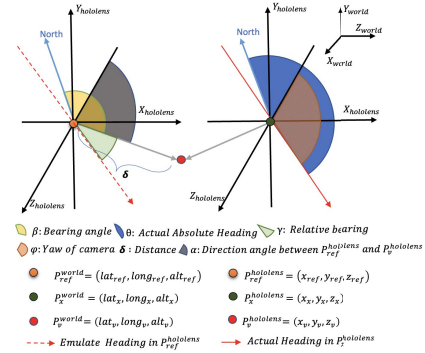


Figure 3: Finding the virtual object position $P_v^{hololens}$ using reference position $P_{ref}^{hololens}$ and the angle α described in the equation 3

4 CONCLUSIONS AND FUTURE WORK

The paper describes an AR application which: 1) Provides safety information for radiological incident. 2) Uses a hybrid GPS and SLAM outdoor framework. 3) Uses KML data to provide real-time information. It provides a way to improve the safety for operatives during such an event. Future work includes improving the user interface and more thorough user testing.

ACKNOWLEDGMENTS

This work was supported by LIST as well as by Luxembourg National Research Fund (RISARX, Grant Number 1534041I)

REFERENCES

- [1] Safety zones. <https://www.epa.gov/emergency-response/safety-zones>. Accessed: 2022-10-26.
- [2] C. R. Z. CRZ. During c&d procedures, biosecurity work zones must be maintained to prevent the spread of microorganisms. as a review, the 3 work zones of a response are the hot zone (exclusion zone), the warm zone (contamination reduction zone), and the cold zone (support zone). 2011.
- [3] G. Klein and D. Murray. Parallel tracking and mapping for small AR workspaces. In *Proc. Sixth IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR'07)*. Nara, Japan, November 2007.
- [4] F. F. Ling, C. Elvezio, J. Bullock, S. Henderson, and S. Feiner. A hybrid rtk gnss and slam outdoor augmented reality system. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 1044-1045, 2019. doi: 10.1109/VR.2019.8798315
- [5] J. E. Swan, D. Hix, and J. L. Gabbard. Perceptual and ergonomic issues in mobile augmented reality for urban operations. 2003.